Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/17/2015
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Project Title:	Exosomes and Secretory Factors as Mediators of Non-targeted Effects of HZE Particles		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
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	This is a proposal devoted to understanding the mechanism of radiation-induced non-targeted effects, and in particular, how they lead to genomic instability in bystander cells (those not directly traversed by a radiation track). Once a bystander cell has become unstable, it is susceptible to mutations induced by endogenous and exogenous stressors, which in turn creates a permissive environment for acquisition of oncogenic mutations. Nontargeted effects (NTE) are a crucial mechanism contributing to solid tumor carcinogenesis in the space environment, yet in many respects they remain poorly understood. We hypothesize that NTE signaling leads to reliance on low-fidelity repair pathways in bystander cells; that the NTE signal is transmitted by exosomes, soluble secretory proteins, or both; and that the NTE signal affects repair through changes in the proteome and transcriptome. To test these hypotheses, we propose three specific aims.		

Task Description:	Aim 1 is to determine effects of radiation quality, dose, and dose rate on genomic instability in bystander cells, as measured by use of low-fidelity repair pathways. Experiments will focus on range of energies for intermediate Z particles. Experiments will use human bronchial epithelial cells, believed to be the target for lung carcinogenesis and bladder cancer cells, used as a positive control because they are known to generate strong non-targeted effect signals. Aim 2 is to determine the non-targeted effect signaling mechanism, with a focus on exosomes and secreted soluble proteins as candidates. Aim 3 is to investigate the exact mechanism by which NTE signaling produces genome instability in bystander cells. In summary, work will improve our quantitative understanding of radiation-induced genome instability, a major mechanism contributing to solid cancer risk, and specifically determine how genomic instability arises as a nontargeted effect in cells not directly traversed by a radiation track.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a project to characterize non-targeted effects of space radiation; that is, effects that are experienced by cells that are not directly traversed by a radiation track. These include many of the same phenomena seen in directly irradiated cells, including activation of the DNA damage response and the induction of mutations and genome instability. Non-targeted effects are thought to amplify the health risks of space radiation exposure, so it is important to know how these effects are transmitted and whether transmission can be blocked, perhaps by pharmacological interventions. One of the novel aspects of the project is to investigate the role of exosomes in transmission of non-targeted effects. Exosomes are nanometer scale, membrane-bounded vesicles capable of carrying information between cells in the form of proteins and RNA. Radiation exposure stimulates exosome production, and one of the goals of the project will be to identify whether space radiation-induced exosomes are different, and perhaps more potent, than terrestrial (low linear energy transfer) radiation-induced exosomes. Although non-targeted effects are harmful in the context of space radiation exposure, they can be beneficial in the context of cancer therapy, where the goal is to do as much damage as possible to tumor cells. Notably, exosomes may provide a mechanism by which dying tumor cells can stimulate destruction of cancer cells that have spread to areas of the body that are outside the scope of the treatment beam. Thus, in addition to helping us to understand how exosomes increase the risks of space radiation, the work may also provide insights into how exosomes might function to make radiation therapy more effective, particularly for advanced cancer that has already started to spread.
Task Progress:	Non-targeted radiation effects are defined as those induced in cells that are not directly traversed by a radiation track. They include many of the same phenomena seen in directly irradiated cells, including activation of the DNA damage response and the induction of mutations and genome instability. Prior work from the investigators showed that non-targeted effects of high charge and energy (HZE) particle radiation arise, in part, from a breakdown in fidelity of DNA repair, as measured by a propensity to misrepair new DNA double-strand breaks introduced at various times following the initial HZE particle exposure. This project focuses on the mechanism of transmission of this non-targeted effect on DNA repair fidelity. The goals of the project are to define the effects of cell type, radiation quality, dose, and dose rate on transmission of the non-targeted effect, with an emphasis on intermediate Z ions at different energies; to determine whether secreted, RNA and protein-containing microvesicles known as exosomes are responsible for transmission non-targeted effect, and to identify the underlying mechanism by which exosomes or other factors evoke the effect. In this first year of the project, experiments were performed according to plan, although the main results will not be available until the end of year one or the early part of year two. Work performed included: - Completion and publication of a manuscript describing the initial finding that breakdown of DNA repair fidelity occurs as a non-targeted effect in radiation-naive cells that have been co-cultured with HZE-irradiated cells. The Principal Investigator (PI) also presented the work in a talk at the May 2015 Space Radiation and Heavy Ions in Therapy Symposium (Osaka), and postdoctoral trainees associated with the project presented posters at the January 2015 NASA Human Research Program Workshop, at the May 2015 International Congress of Radiation Research (Kyoto), and at the September 2015 Radiation Research Society annual meeting (Weston, FL). - Establishment of a new assay system for repair fidelity, based on mis-joining of double-strand breaks in endogenous genes (rather than the artificial reporter genes used in earlier work). The assay uses the CRISPR/Cas9 nuclease to induce a paracentric inversion involving the EML4 and ALK genes. This is the most commonly occurring rearrangement in lung cancer patients and is particularly frequent in never-smokers, the population of most interest to NASA. A sensitive and quantitative qPCR assay was established to measure the relative efficiency of the CRISPR/Cas9-induced inversion in a number of candidate cell lines. Three of these cell lines were chosen for radiation studies: near-normal human bronchial epithelial cells (HBEC3-KT); HZE radiation-transformed HBEC3-KT cells (HBECK3KT-F25F); and, as a positive control, the human bladder cancer cells (DsR-7F4) used in previous work by the investigators. - Irradiation of a panel of human cell lines in the NASA Space Radiation Laboratory (NSRL) 15C (Fall, 2015) run. To define radiation quality effects, 16-O, 28-Si, and 48-Ti ions were used at energies calculated to provide linear energy transfer (LET) values of 100 and 200 keV/μm. These represent a range of species that might be encountered within a shielded crew habitat. Irradiation was performed at doses ranging form 0.2 to 1.0 Gy. Initial results showed that radiation led to a modest, dose-dependent reduction in cell growth rate, as expected. At day 7 post-irradiation, the directly irradiated and bystander cell populations were challenged with CRISPR/Cas9 nuclease to induce targeted strand breaks in the EML4 and ALK loci. Molecular analysis of the relative frequency of inversions in different treatment groups is in progress and is expected to be complete, or nearly complete, by the end of year one of the award (January 2016). - Establishment of methods for analysis of radiation-induced exosomes, including light scattering and electron microscopy to determine the size distribution, immunoblotting, and mass spectrometry to identify exosomal proteins, and fluorescence labeling to allow visualization of uptake in bystander cells. Irradiated cell conditioned medium was also collected from 48-Ti-irradiated cells in the NSRL 15A and 15C runs (Spring and Fall 2015). This conditioned medium will be used to prepare exosomes for proteomic analysis. Initial results will be available in early 2016.

Bibliography Type:	Description: (Last Updated: 08/06/2025)
Articles in Peer-reviewed Journals	Li Z, Doho G, Zheng X, Jella KK, Li S, Wang Y, Dynan WS. "Co-culturing with high-charge and energy particle irradiated cells increases mutagenic joining of enzymatically induced DNA double-strand breaks in nonirradiated cells." Radiat Res. 2015 Sep;184(3):249-58. http://dx.doi.org/10.1667/RR14092.1 ; PubMed PMID: 26284422 , Sep-2015